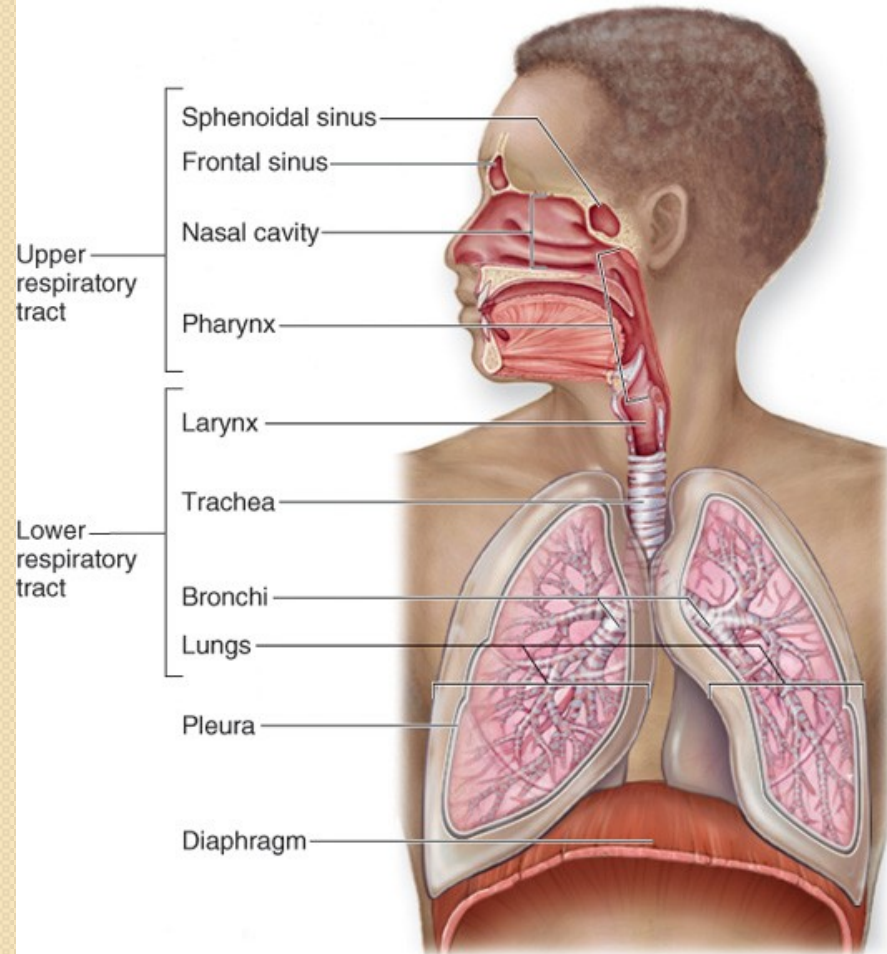


RESPIRATORY SYSTEM

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.

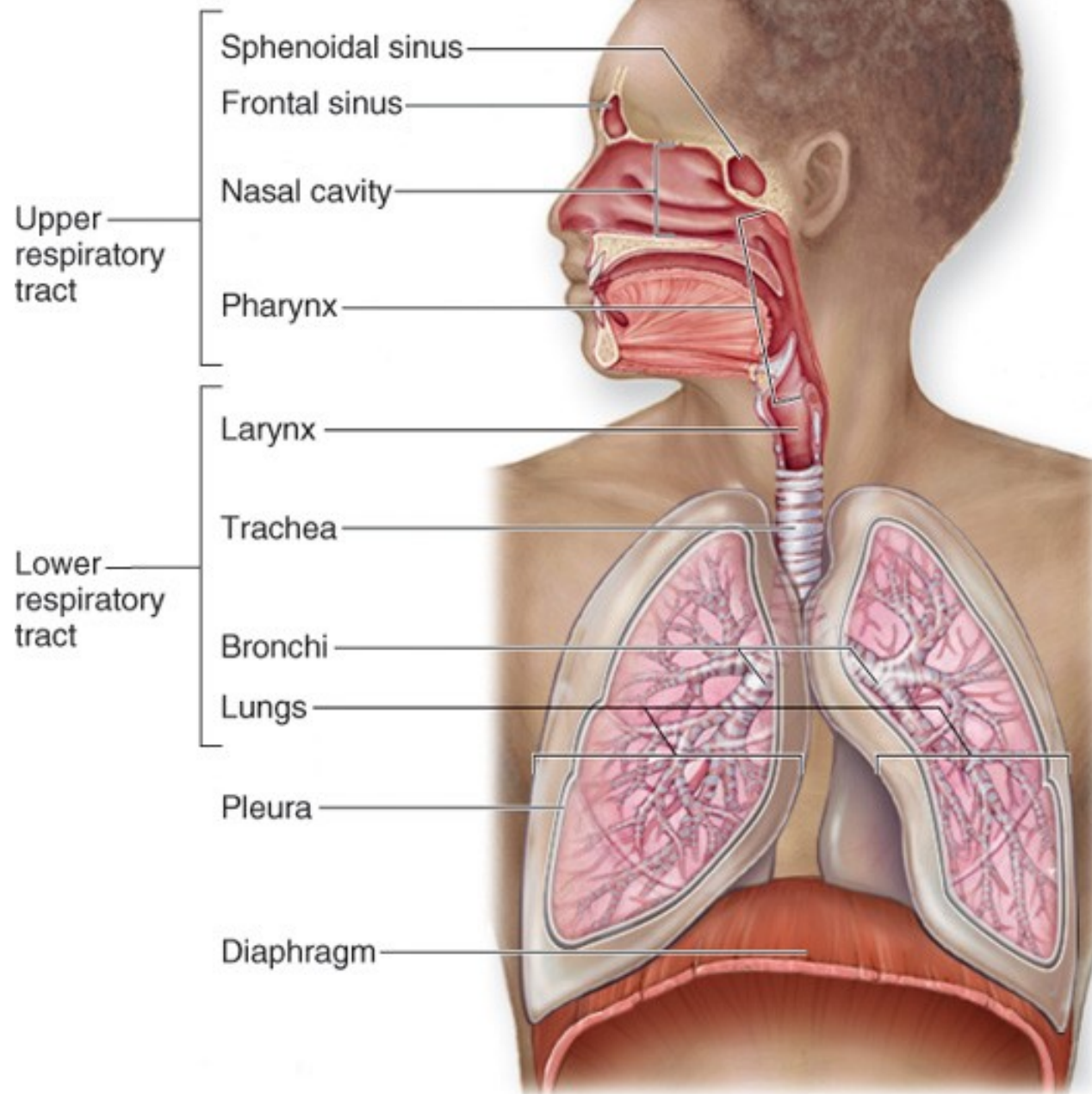


Organization and Functions of the Respiratory System

- Structural classifications:
 - **upper respiratory tract**
 - **lower respiratory tract.**
- Functional classifications:
 - **Conducting** portion: transports air.
 - Nose
 - nasal cavity
 - Pharynx
 - Larynx
 - Trachea
 - progressively smaller airways, from the primary bronchi to the bronchioles

Organization and Functions of the Respiratory System

- Functional classifications: continued
 - **Conducting** portion: transports air.
 - **Respiratory** portion: carries out gas exchange.
 - respiratory bronchioles
 - alveolar ducts
 - air sacs called alveoli
- Upper respiratory tract is all conducting
- Lower respiratory tract has both conducting and respiratory portions



Respiratory System Functions

- **Breathing (pulmonary ventilation):**
 - consists of **two cyclic phases**:
 - **inhalation**, also called **inspiration**
 - **exhalation**, also called **expiration**
 - Inhalation draws gases **into** the lungs.
 - Exhalation forces gases **out** of the lungs.
- **Gas exchange: O_2 and CO_2**
 - External respiration
 - External environment and blood
 - Internal respiration
 - Blood and cells

Respiratory System Functions

- **Gas conditioning:**
 - Warmed
 - Humidified
 - Cleaned of particulates
- **Sound production:**
 - Movement of air over true vocal cords
 - Also involves nose, paranasal sinuses, teeth, lips and tongue
- **Olfaction:**
 - Olfactory epithelium over superior nasal conchae
- **Defense:**
 - Course hairs, mucus, lymphoid tissue

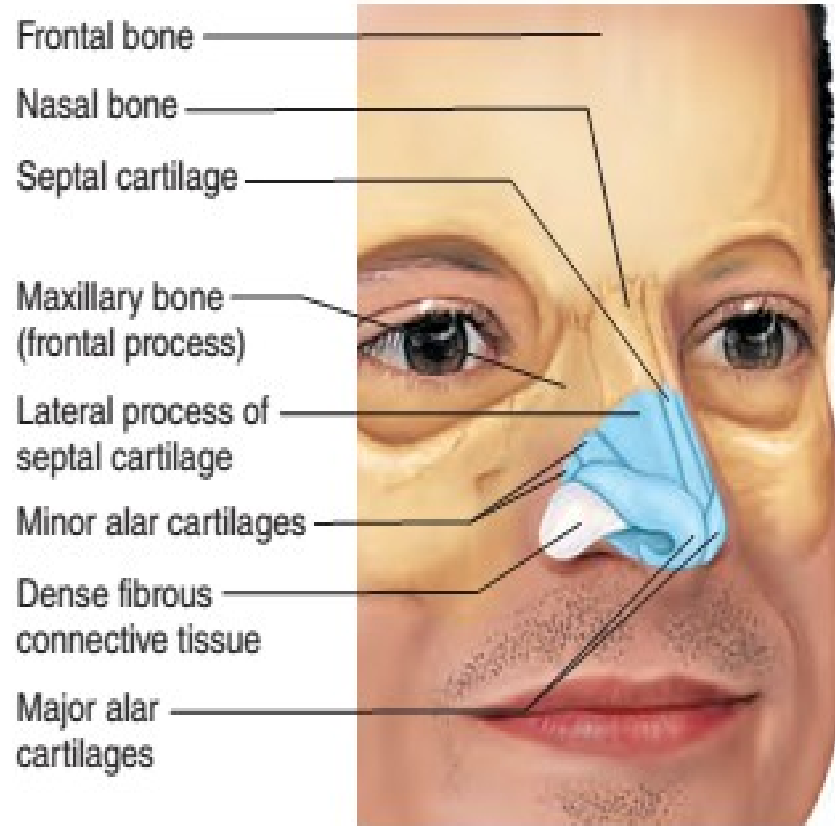
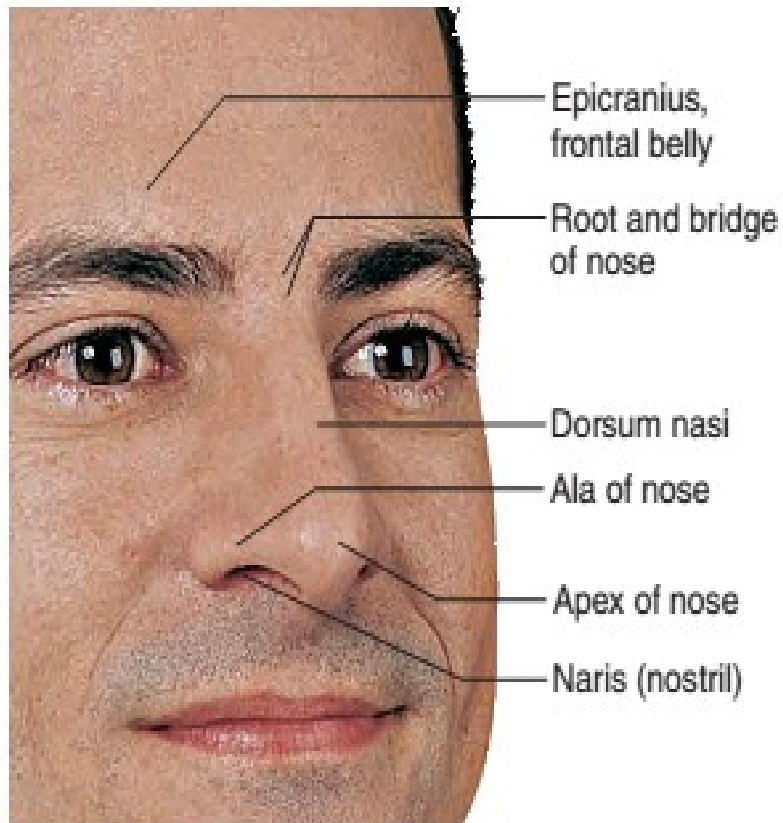
Upper Respiratory Tract

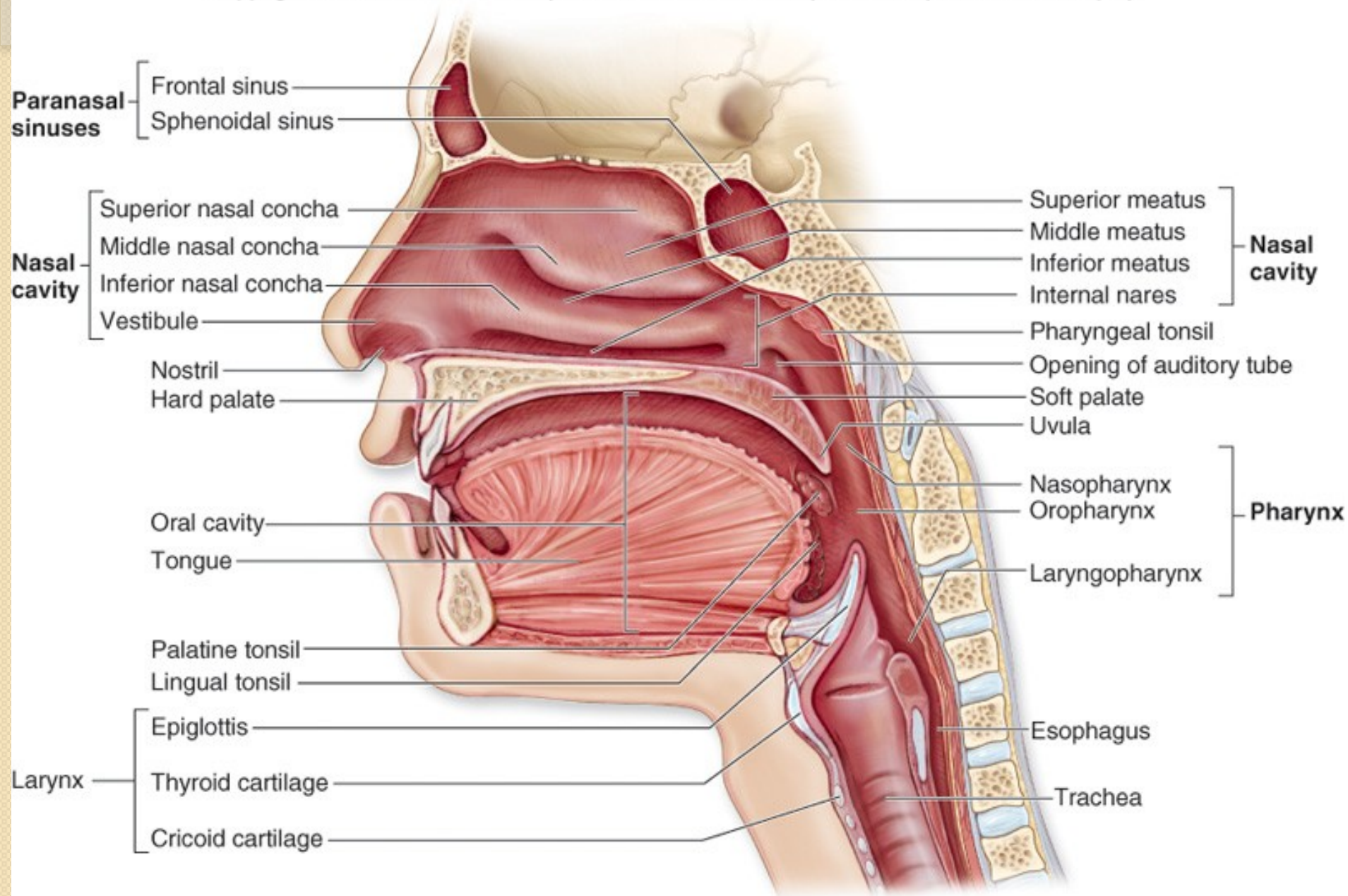
- Composed of
 - nose
 - nasal cavity
 - paranasal sinuses
 - pharynx (throat)
- All are part of the **conducting** portion of the respiratory system.

Respiratory System Organization

- **The Nose**

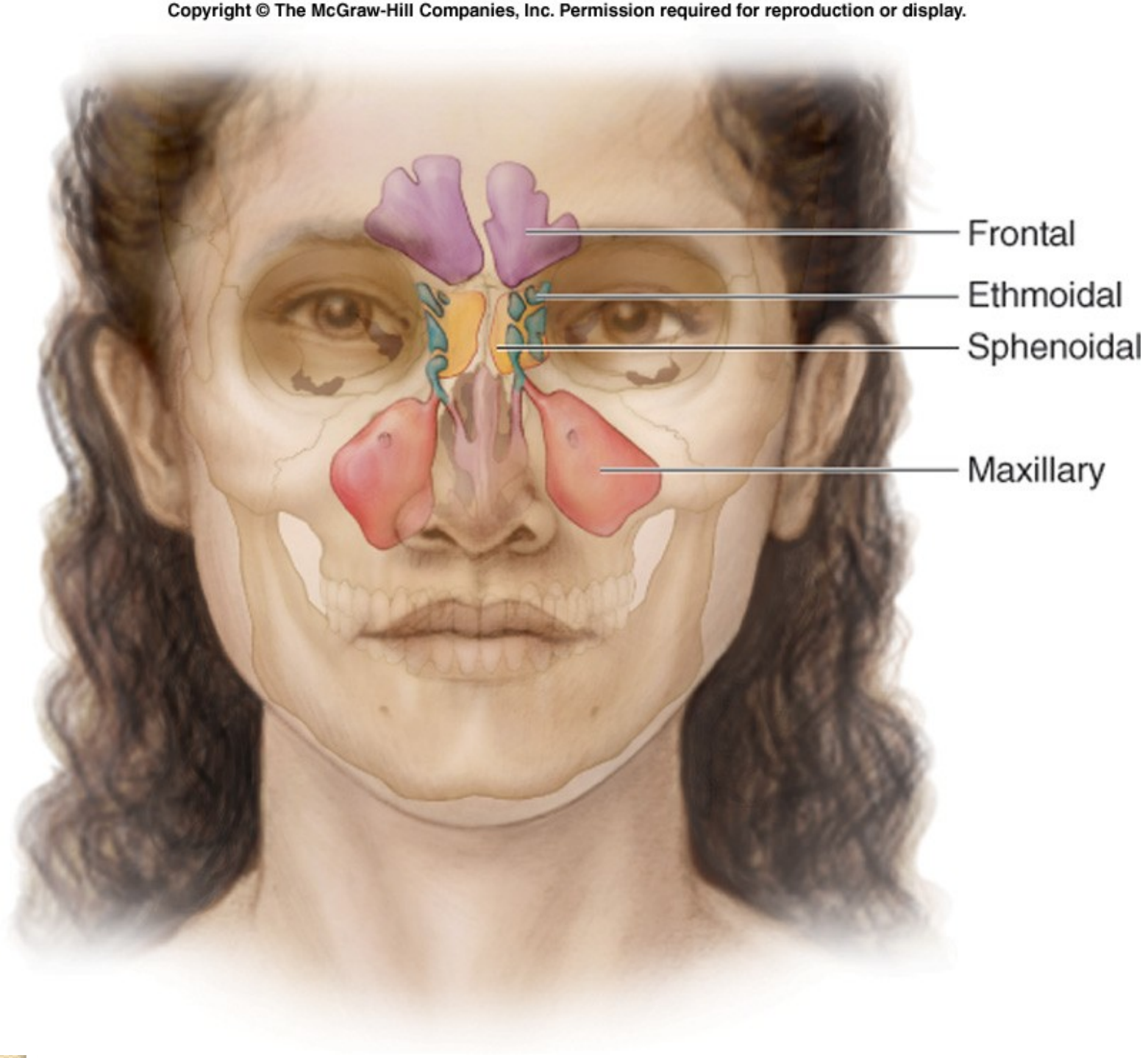
- **External nares (nostrils)** admit air
 - Nasal vestibule lined with hairs to filter air
- **Vestibule** opens into **nasal cavity**
 - Hard palate separates nasal and oral cavities
- Cavity continues through internal nares to nasopharynx
 - Soft palate underlies nasopharynx





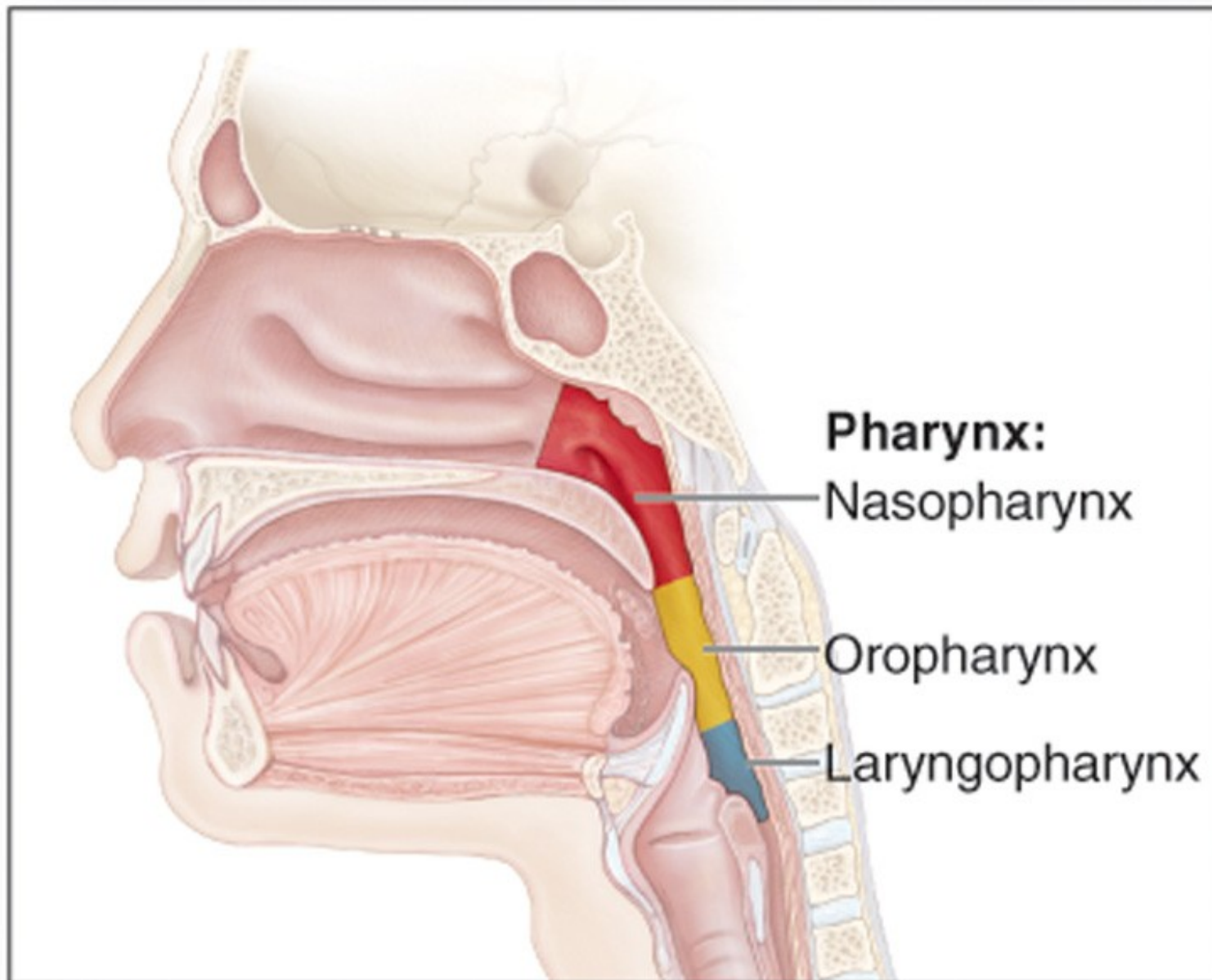
Paranasal Sinuses

- **Paranasal sinuses:**
 - In four skull bones
 - paired air spaces
 - **decrease** skull bone weight
- Named for the bones in which they are housed.
 - **frontal**
 - **ethmoidal**
 - **sphenoidal**
 - **maxillary**
- Communicate with the nasal cavity by ducts.



Pharynx

- Common to both the respiratory and digestive systems.
- Commonly called the **throat**.
- Funnel-shaped
 - slightly wider superiorly and narrower inferiorly.
- Originates posterior to the nasal and oral cavities
- Extends inferiorly near the level of the bifurcation of the larynx and esophagus.
- Common pathway for both air and food.



(b) Regions of pharynx

Pharynx

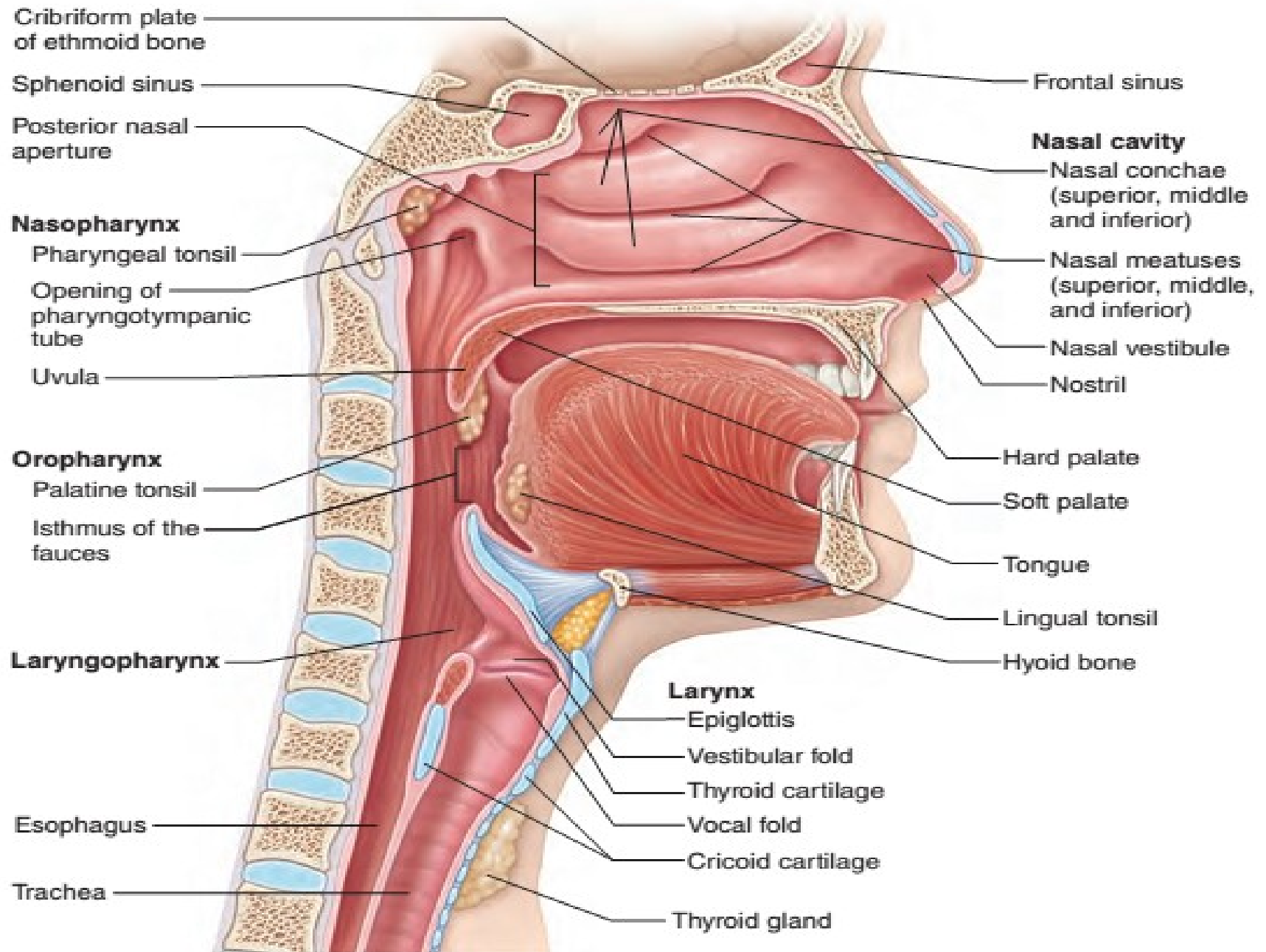
- **Walls:**
 - lined by a mucosa
 - contain skeletal muscles primarily used for **swallowing**.
- Flexible lateral walls
 - distensible
 - to force swallowed food into the esophagus.
- Partitioned into three adjoining regions:
 - **nasopharynx**
 - **oropharynx**
 - **laryngopharynx**

Nasopharynx

- **Superiormost region of the pharynx.**
- Location:
 - posterior to the nasal cavity
 - superior to the soft palate
 - separates it from the posterior part of the oral cavity.
- Normally, only air passes through.
- **Soft palate**
 - Blocks material from the oral cavity and oropharynx
 - elevates when we swallow.

Nasopharynx

- **Auditory tubes**
 - paired
 - In the lateral walls of the nasopharynx
 - connect the nasopharynx to the middle ear.
- **Pharyngeal tonsil**
 - posterior nasopharynx wall
 - single
 - commonly called the adenoids.



Oropharynx

- The middle pharyngeal region.
- Location:
 - Immediately posterior to the oral cavity.
 - Bounded by the soft palate superiorly,
 - the hyoid bone inferiorly.
- Common respiratory and digestive pathway
 - both air and swallowed food and drink pass through.
- 2 pairs of muscular arches
 - **anterior palatoglossal arches**
 - **posterior palatopharyngeal arches**
 - form the entrance from the oral cavity.

Oropharynx

- Lymphatic organs
 - provide the “first line of defense” against ingested or inhaled foreign materials.
 - **Palatine tonsils**
 - on the lateral wall between the arches
 - **Lingual tonsils**
 - At the base of the tongue.

Laryngopharynx

- Inferior, narrowed region of the pharynx.
- Location:
 - Extends inferiorly from the hyoid bone
 - is continuous with the larynx and esophagus.
 - Terminates at the superior border of the esophagus
 - is equivalent to the inferior border of the cricoid cartilage in the larynx.
- The **larynx** (voice box) forms the anterior wall
- Permits passage of both food and air.

Lower Respiratory Tract

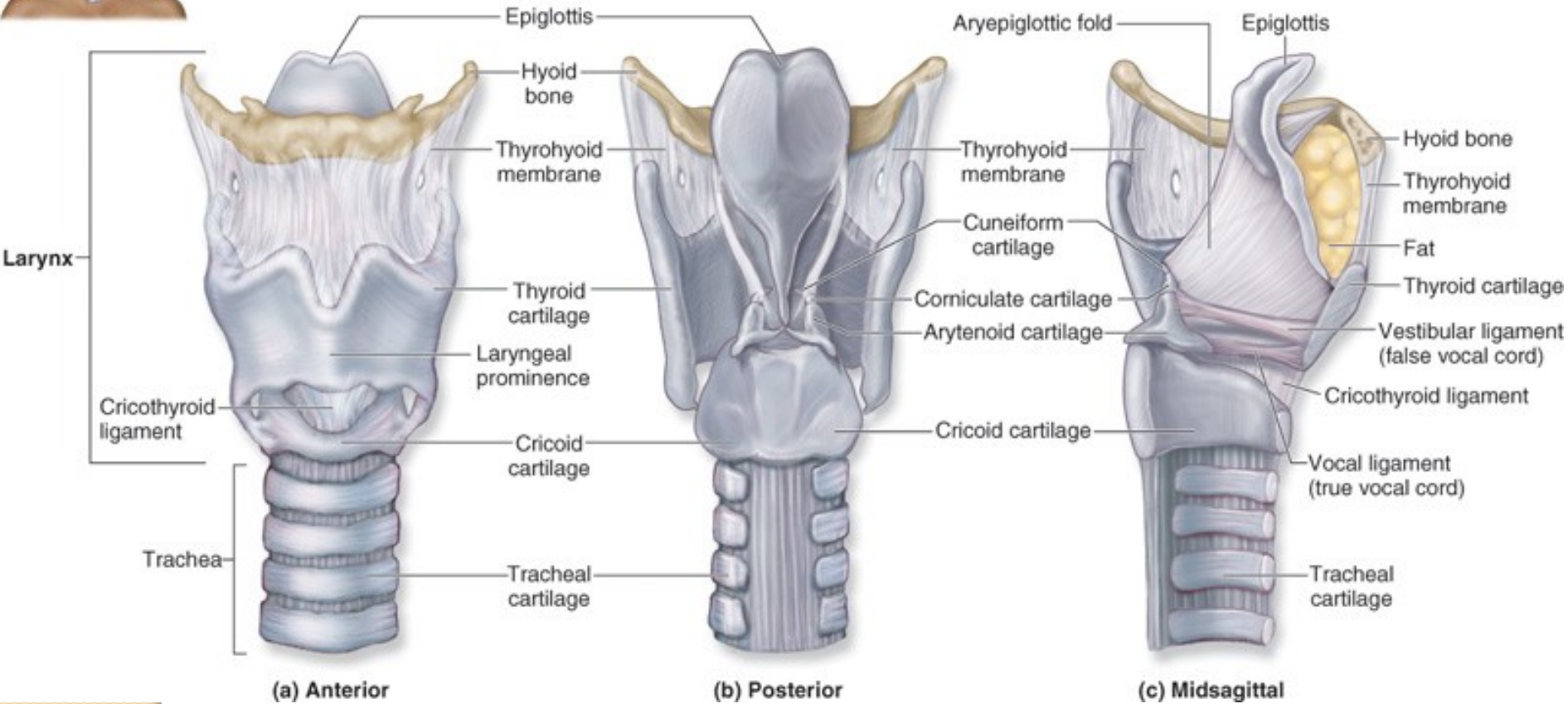
- **Conducting portion**
 - Larynx
 - Trachea
 - Bronchi
 - bronchioles and their associated structures
- **Respiratory portion** of the respiratory system
 - respiratory bronchioles
 - alveolar ducts
 - alveoli

Larynx

- Short, somewhat cylindrical airway
- Location:
 - bounded posteriorly by the laryngopharynx,
 - inferiorly by the trachea.
- Prevents swallowed materials from entering the lower respiratory tract.
- Conducts air into the lower respiratory tract.
- Produces sounds.

Larynx

- **Nine** pieces of cartilage
 - three individual pieces
 - Thyroid cartilage
 - Cricoid cartilage
 - Epiglottis
 - three cartilage pairs
 - Arytenoids: on cricoid
 - Corniculates: attach to arytenoids
 - Cuneiforms: in aryepiglottic fold
 - held in place by ligaments and muscles.
 - Intrinsic muscles: regulate tension on true vocal cords
 - Extrinsic muscles: stabilize the larynx

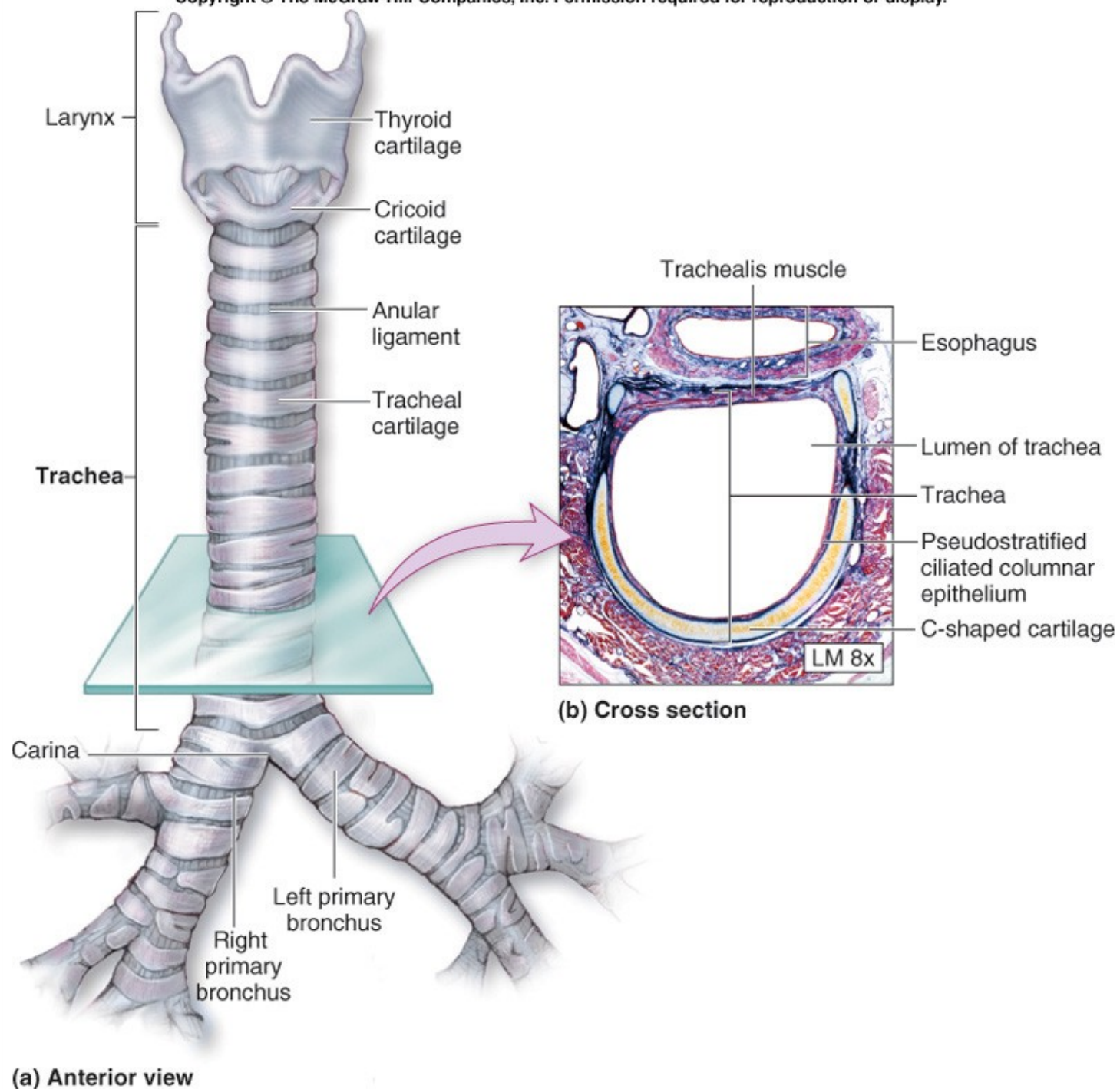


Larynx

- **Thyroid cartilage**
 - Largest hyaline cartilage
 - Protrudes anteriorly (Adam's apple)
- **Epiglottis**
 - Superior opening of the larynx
 - Protects larynx during swallowing
- *False vocal cords*
 - Act as valves
- *True vocal cords* (vocal folds)
 - Vibrate with expelled air to create sound (speech)
- *Glottis* – opening between vocal cords

Trachea

- A flexible, slightly rigid tubular organ
 - often referred to as the “windpipe.”
- Extends through the mediastinum
 - immediately anterior to the esophagus
 - inferior to the larynx
 - superior to the primary bronchi of the lungs.
- Anterior and lateral walls of the trachea are supported by 15 to 20 C-shaped tracheal cartilages.
 - cartilage rings reinforce and provide some rigidity to the tracheal wall to ensure that the trachea remains open (patent) at all times
 - cartilage rings are connected by elastic sheets called anular ligaments



Trachea

- At the level of the **sternal angle**, the trachea bifurcates into two smaller tubes, called the **right and left primary bronchi**.
- Each primary bronchus projects laterally toward each lung.
- The most inferior tracheal cartilage separates the primary bronchi at their origin and forms an internal ridge called the **carina**.

Bronchial Tree

- A highly branched system
 - air-conducting passages
 - originate from the left and right primary bronchi.
- Progressively branch into narrower tubes as they diverge throughout the lungs before terminating in terminal bronchioles.
- **Primary bronchi**
 - Incomplete rings of hyaline cartilage ensure that they remain open.
 - Right primary bronchus
 - shorter, wider, and more vertically oriented than the left primary bronchus.
 - Foreign particles are more likely to lodge in the right primary bronchus.

Bronchial Tree

- **Primary bronchi**
 - enter the hilum of each lung
 - Also entering hilum:
 - pulmonary vessels
 - lymphatic vessels
 - nerves.
- **Secondary bronchi (or lobar bronchi)**
 - Branch of primary bronchus
 - left lung:
 - two lobes
 - two secondary bronchi
 - right lung
 - three lobes
 - three secondary bronchi.

Bronchial Tree

- Tertiary bronchi (or segmental bronchi)
 - Branch of secondary bronchi
 - left lung is supplied by 8 to 10 tertiary bronchi.
 - right lung is supplied by 10 tertiary bronchi
 - supply a part of the lung called a bronchopulmonary segment



- Primary bronchi
- Secondary bronchi
- Tertiary bronchi
- Smaller bronchi



Right primary bronchus

Right secondary bronchus

Right tertiary bronchus

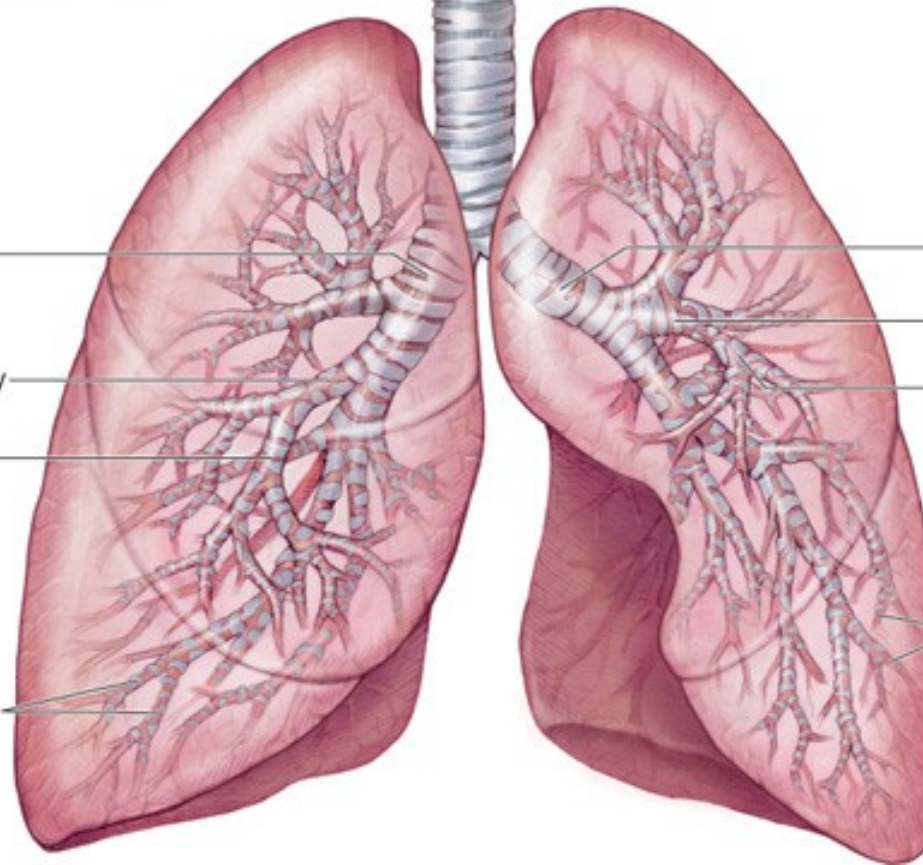
Smaller bronchi

Left primary bronchus

Left secondary bronchus

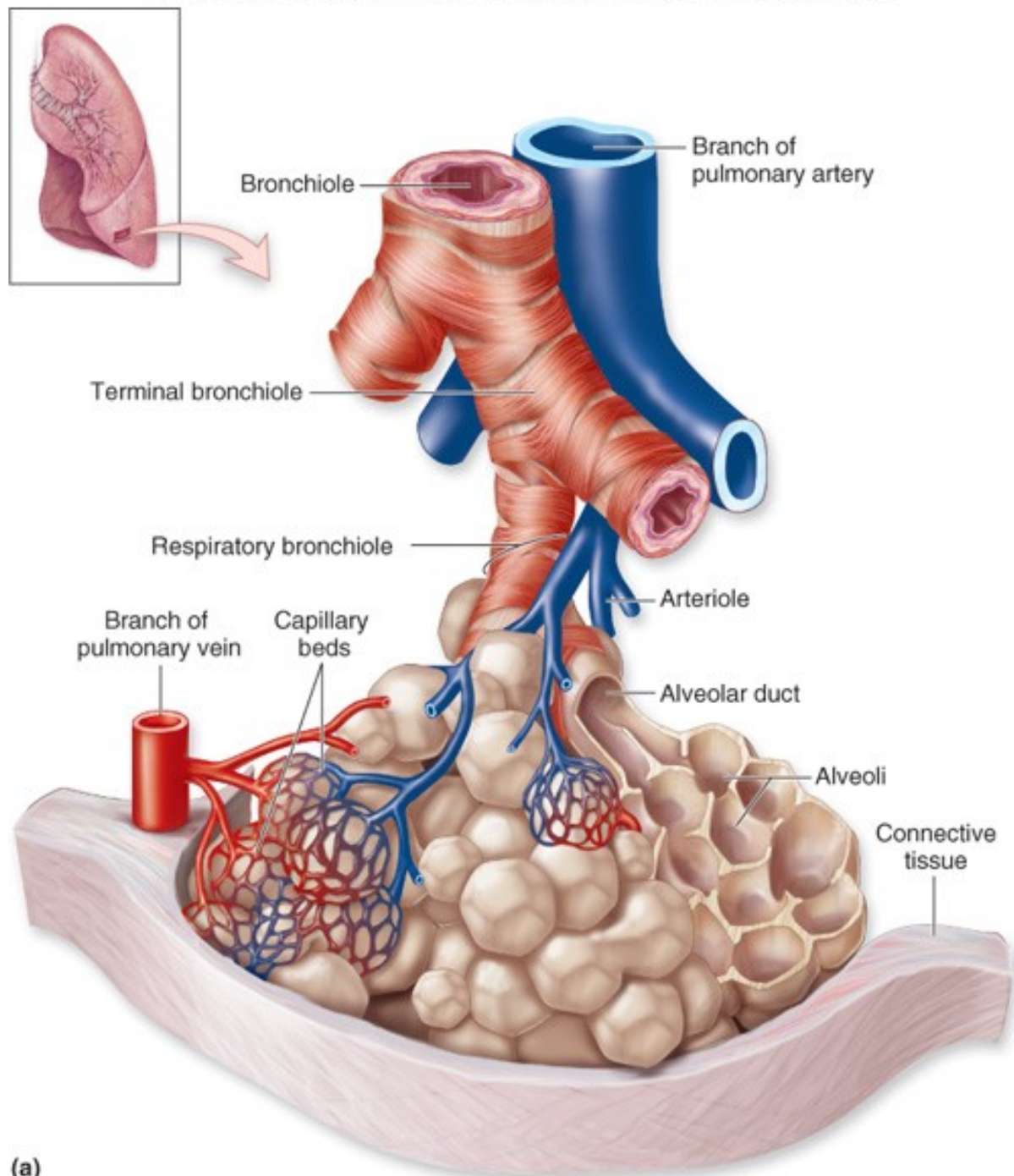
Left tertiary bronchus

Smaller bronchi

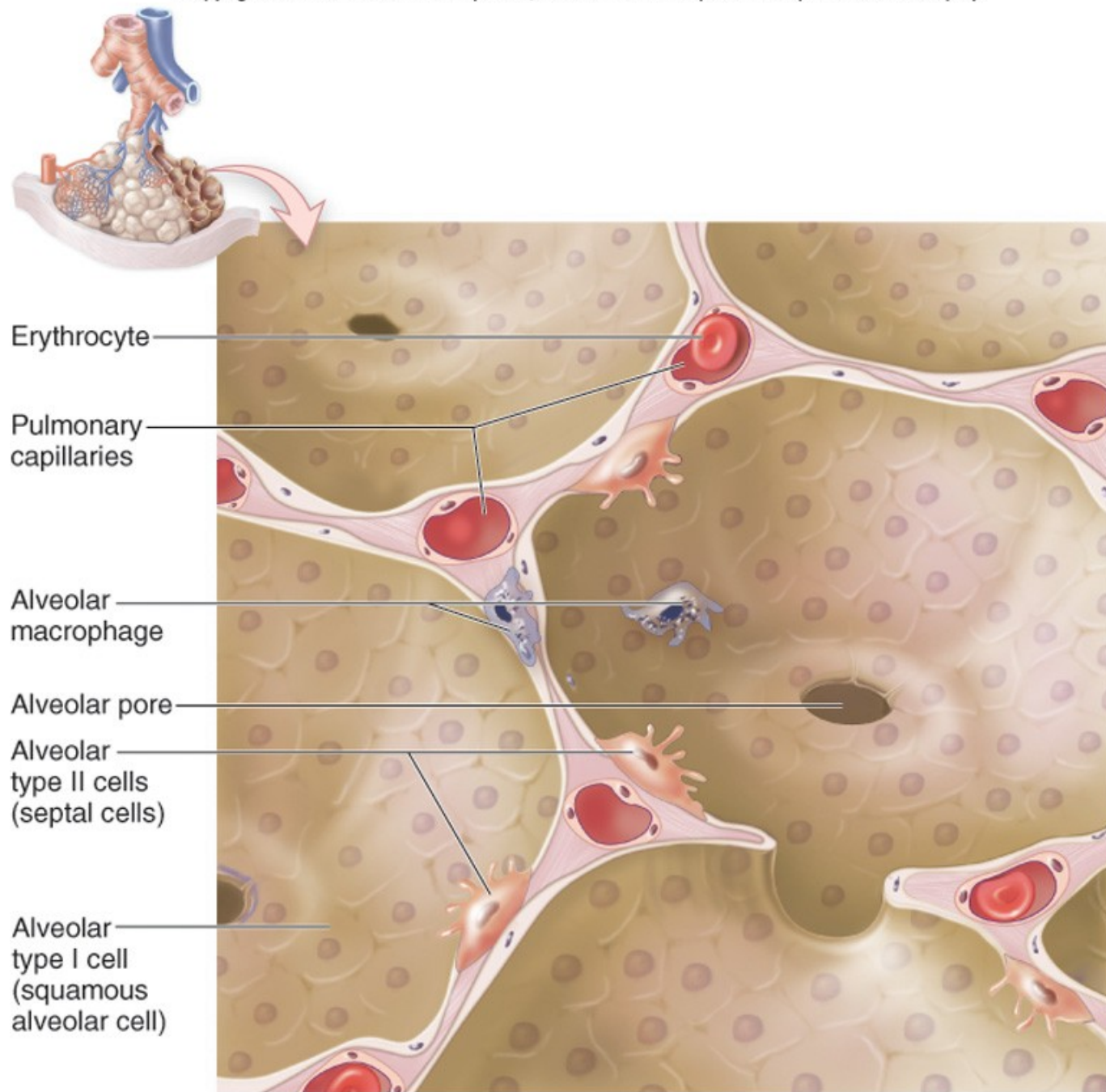


Respiratory Bronchioles, Alveolar Ducts, and Alveoli

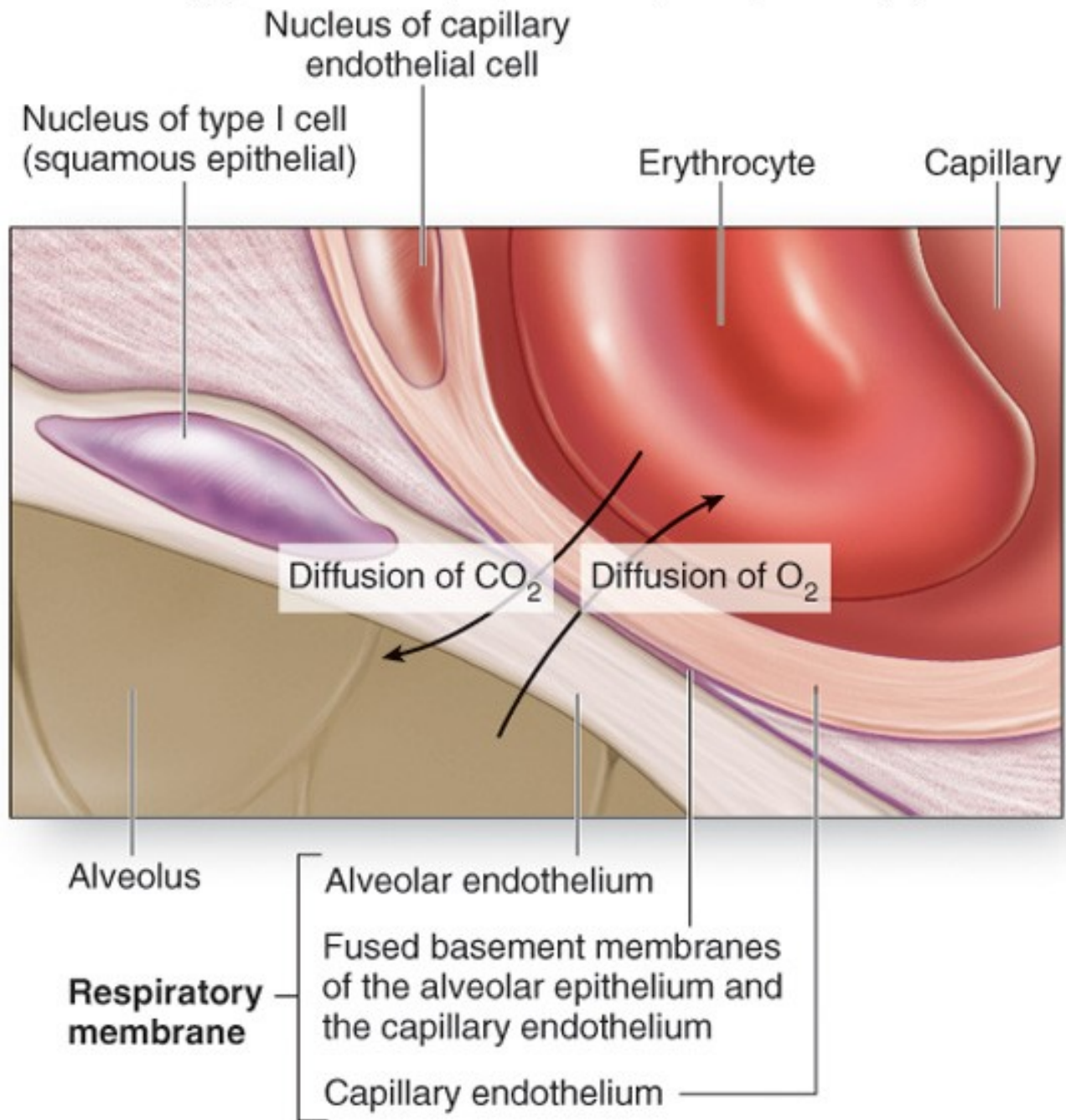
- Contain small saccular outpocketings called **alveoli**.
- An alveolus is about 0.25 to 0.5 millimeter in diameter.
- Its thin wall is specialized to **promote diffusion of gases** between the alveolus and the blood in the pulmonary capillaries.
- Gas exchange can take place in the respiratory bronchioles and alveolar ducts as well as in the lungs, which contain approximately 300–400 million alveoli.
- The **spongy** nature of the lung is due to the packing of millions of alveoli together.



(a)

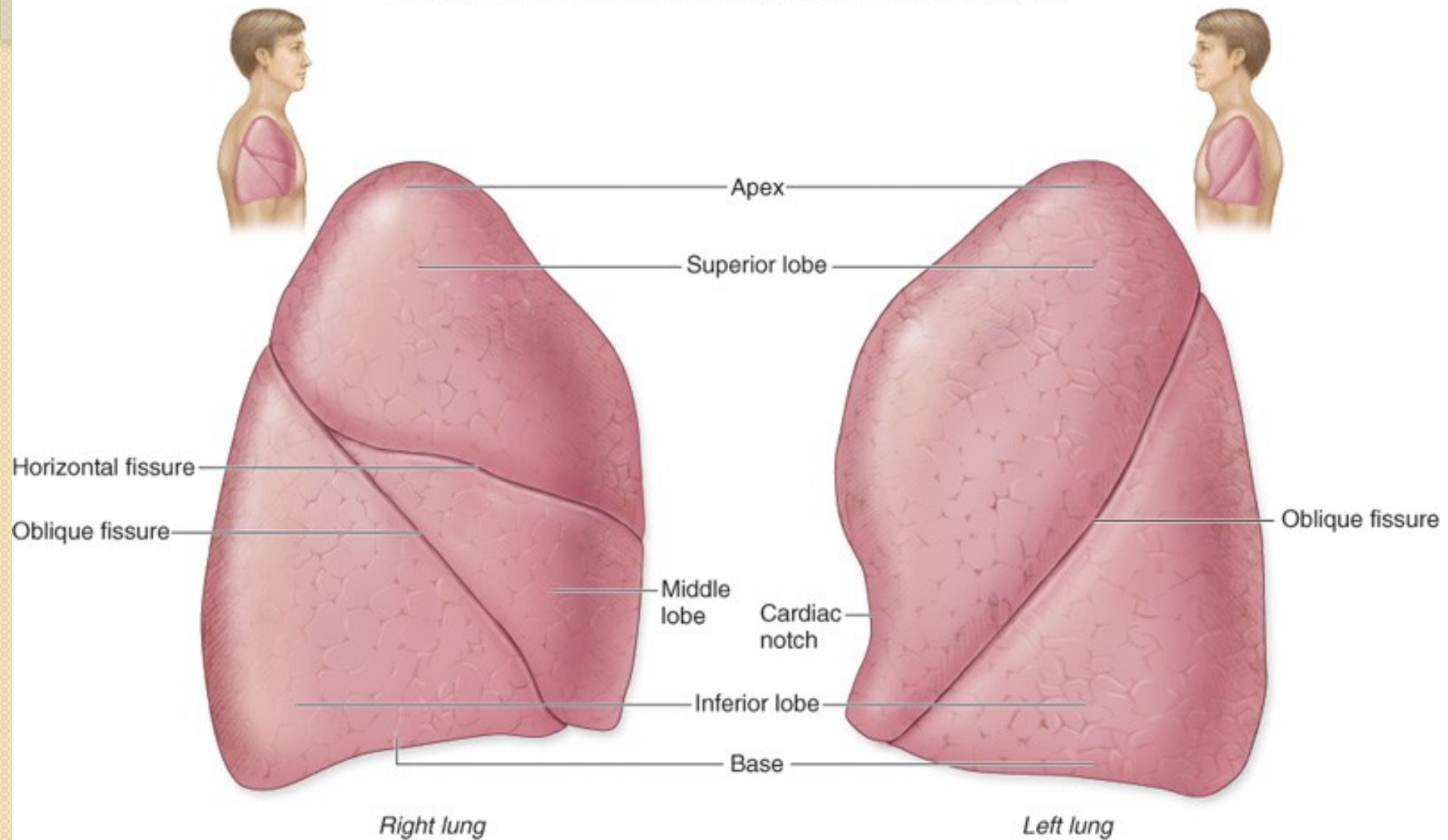


(a)

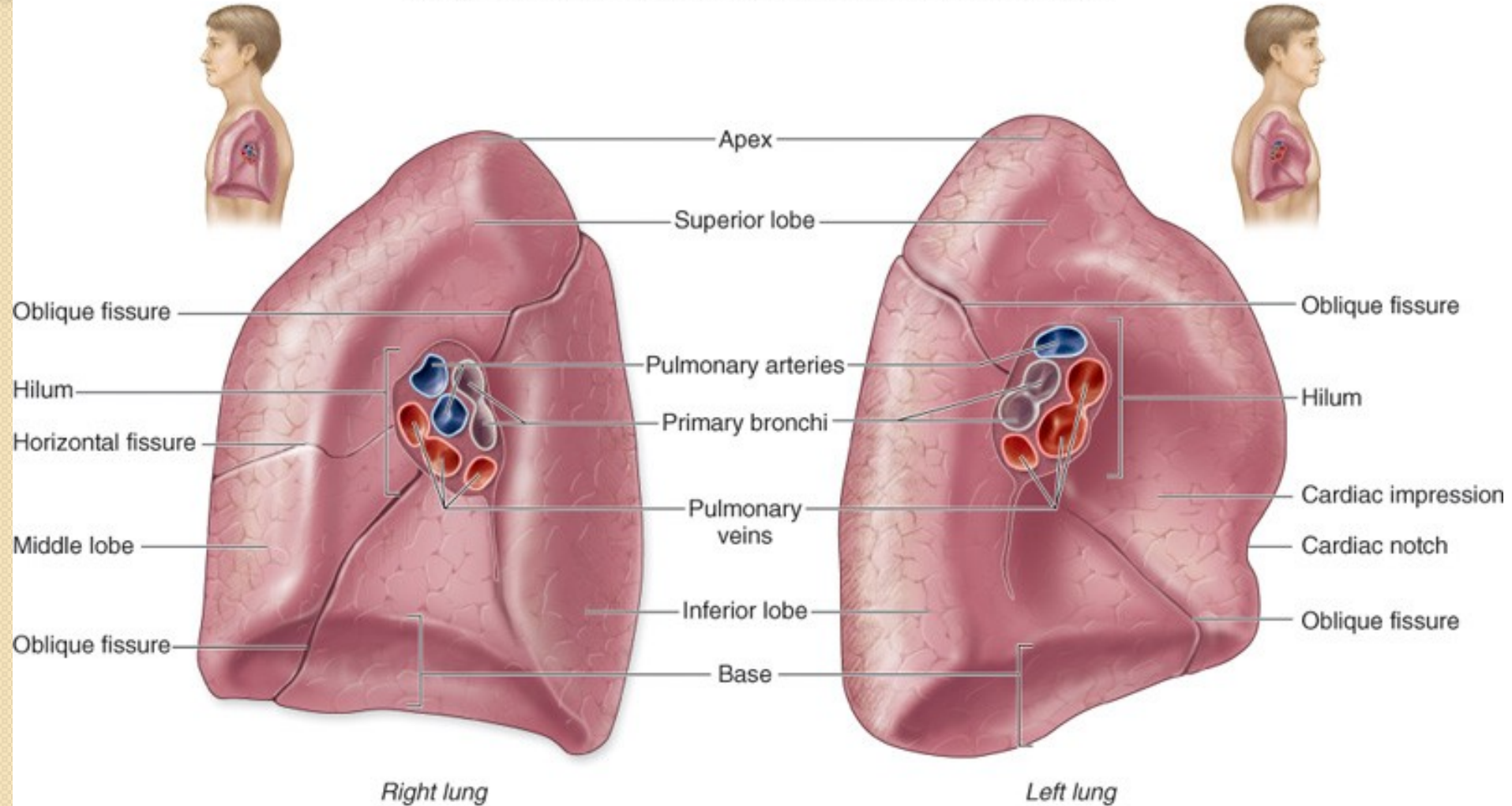


Gross Anatomy of the Lungs

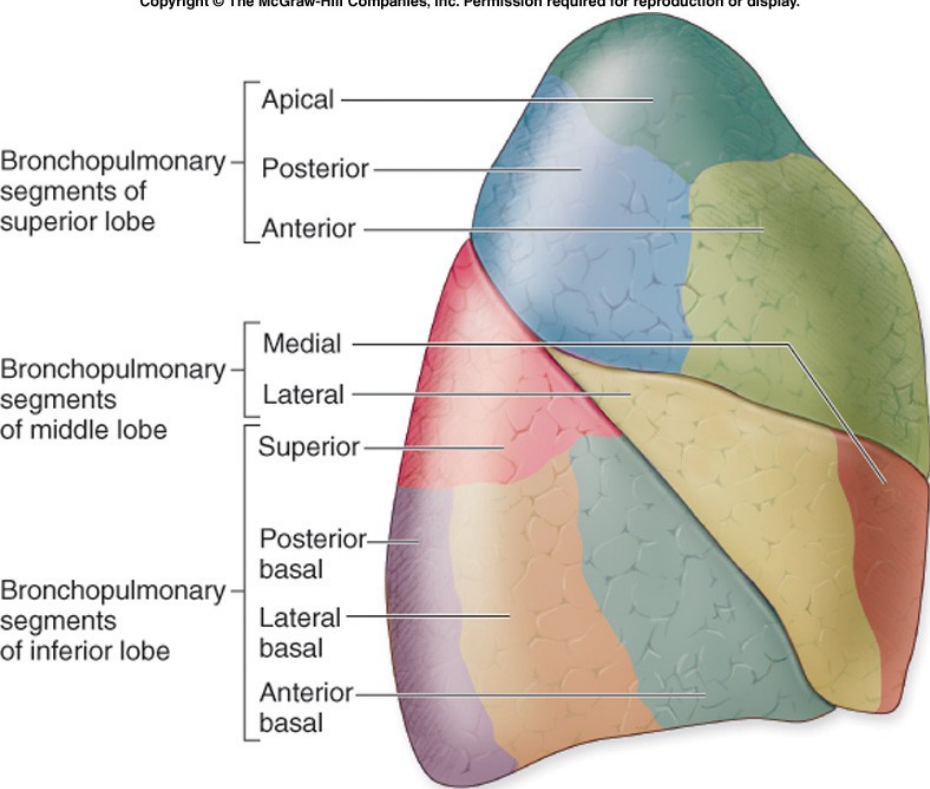
- Each lung has a **conical** shape.
- Its wide, concave **base** rests upon the **muscular diaphragm**.
- Its relatively blunt superior region, called the **apex** or (cupola), projects superiorly to a point that is slightly superior and posterior to the clavicle.
- Both lungs are bordered by the thoracic wall anteriorly, laterally, and posteriorly, and supported by the **rib cage**.
- Toward the midline, the lungs are separated from each other by the **mediastinum**.
- The relatively broad, rounded surface in contact with the thoracic wall is called the **costal surface** of the lung.



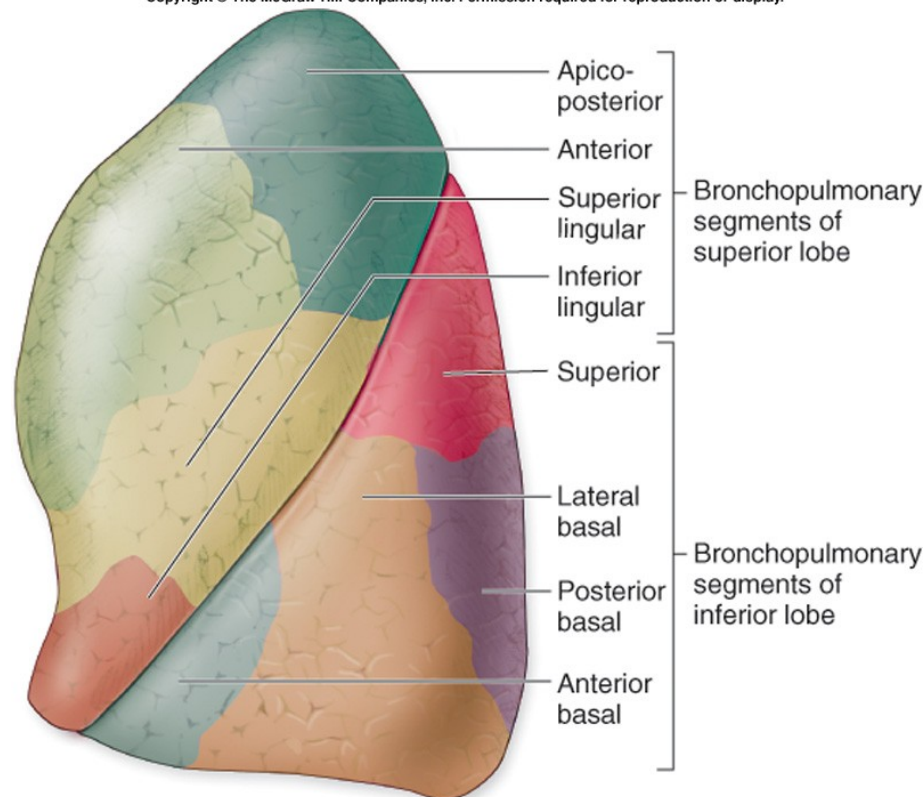
(a) Lateral views



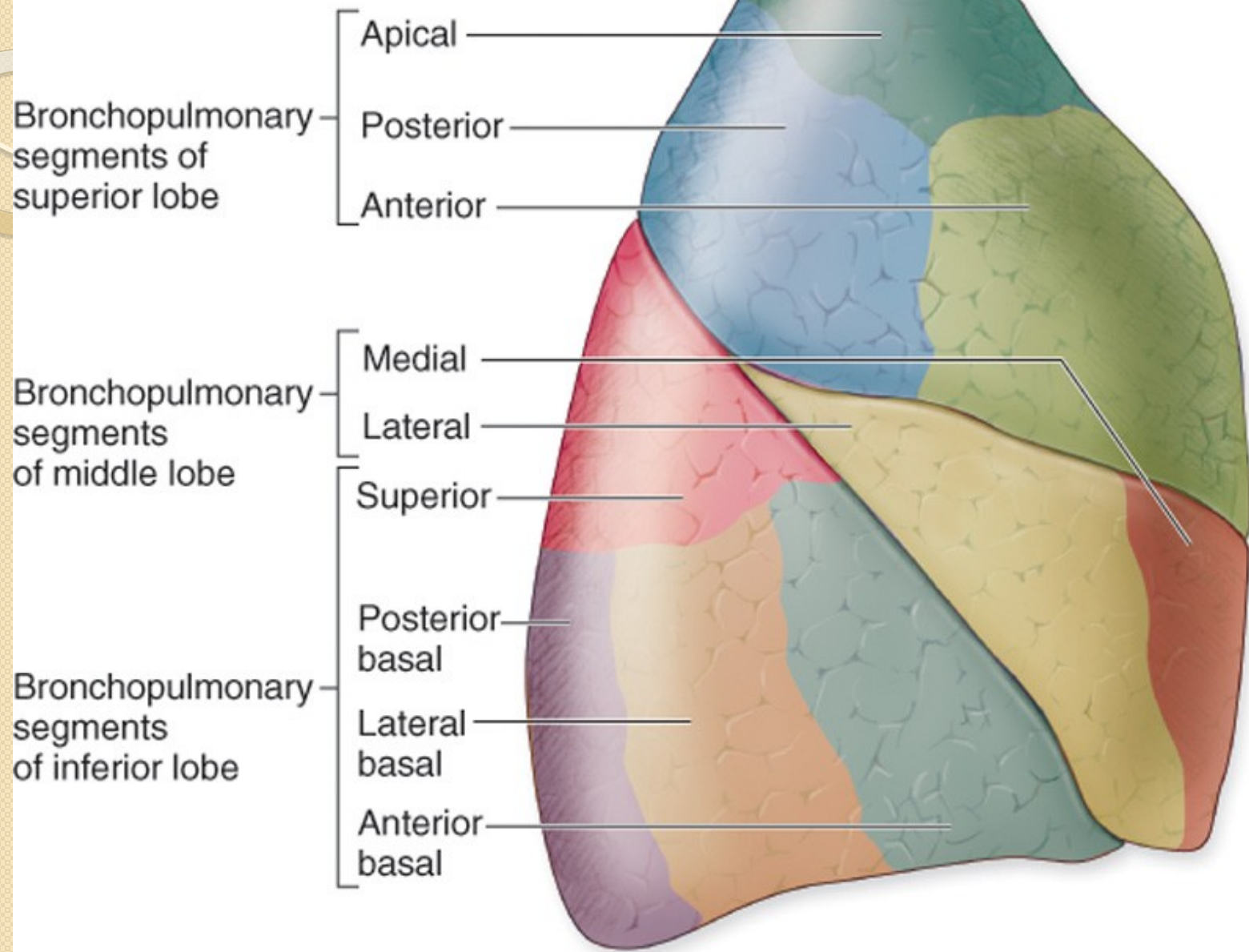
(b) Medial views



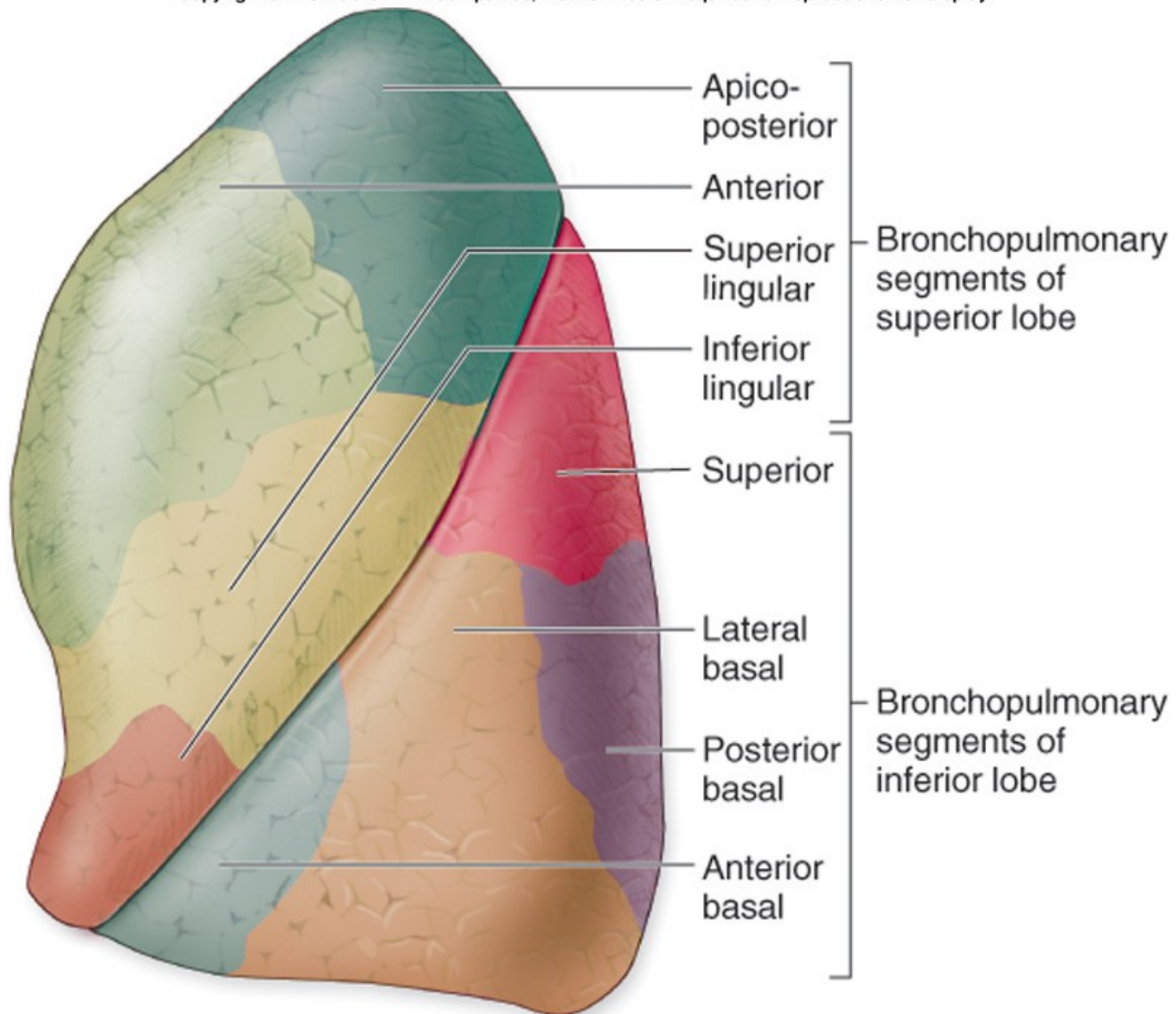
Right lung, lateral view



Left lung, lateral view



Right lung, lateral view



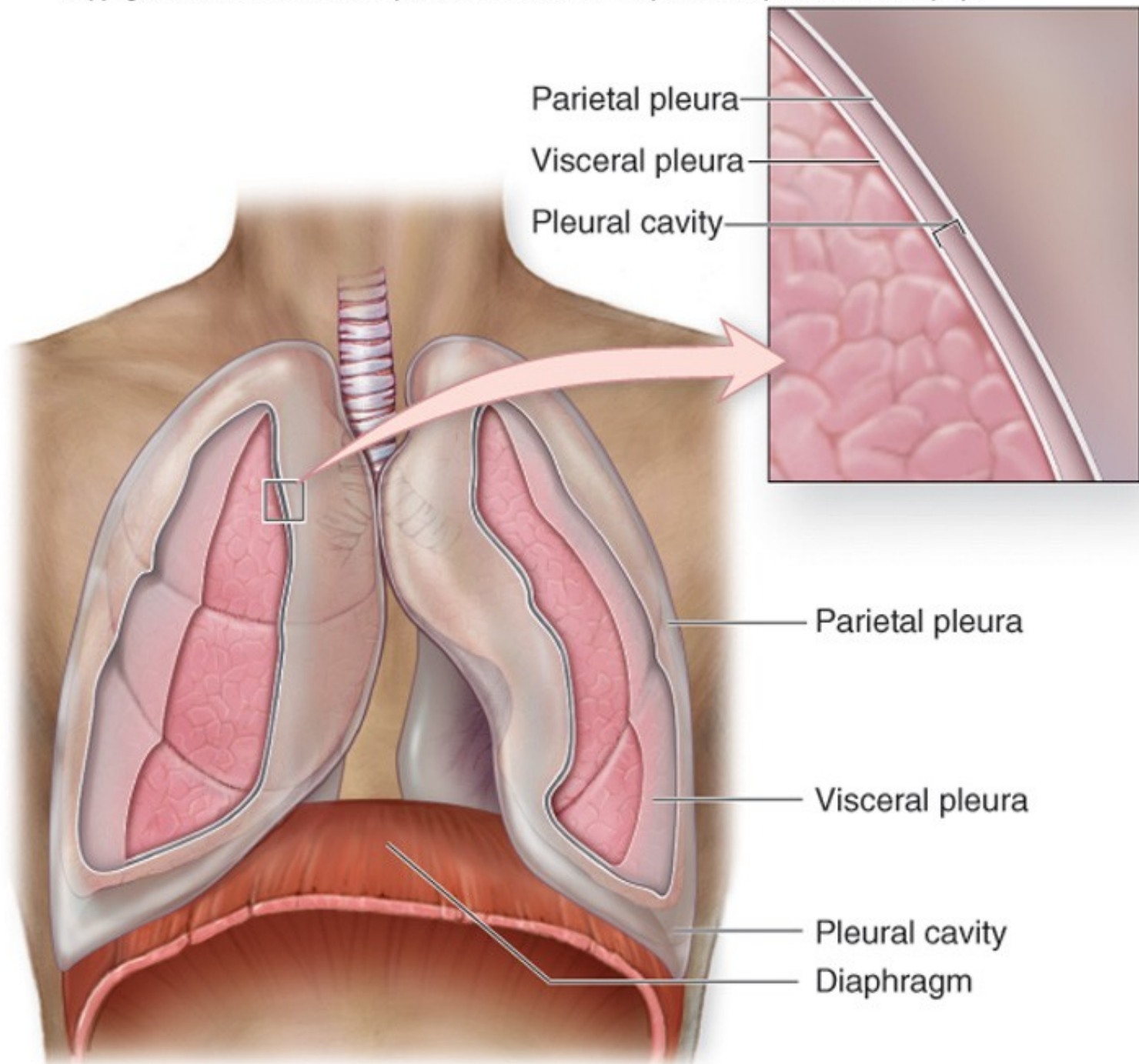
Left lung, lateral view

Pleura and Pleural Cavities

- The outer surface of each lung and the adjacent internal thoracic wall are lined by a serous membrane called **pleura**, which is formed from simple squamous epithelium.
- The outer surface of each lung is tightly covered by the **visceral pleura**, while the internal thoracic walls, the lateral surfaces of the mediastinum, and the superior surface of the diaphragm are lined by the **parietal pleura**.
- The parietal and visceral pleural layers are continuous at the hilum of each lung.
- The **potential space** between these serous membrane layers is a **pleural cavity**.
- The pleural membranes produce a thin, serous **fluid** that circulates in the pleural cavity and acts as a **lubricant**, ensuring minimal friction during breathing.

Pneumothorax

- **Free air in pleural cavity** from an external injury or internally (e.g., broken rib)
- Collapsed lung or part of lung results = atelectasis
- Large amount requires a tube for removal
- Dangerous type is tension pneumothorax where hole acts as a one-way valve
- Fluids may accumulate in pleural cavity
 - Hemothorax if blood
 - Hydrothorax if serous fluid
 - Empyema if pus



Pneumonia

- **Infection in the alveoli**
- Causes: viruses, bacteria, and sometimes fungi
- Tissue swelling and WBCs accumulate decreasing gas exchange
- Contagious
- Symptoms: cough, fever, rapid breathing, sputum
- Treatment: antibiotics, respiratory support, and medications for symptoms and possibly supplemental oxygen

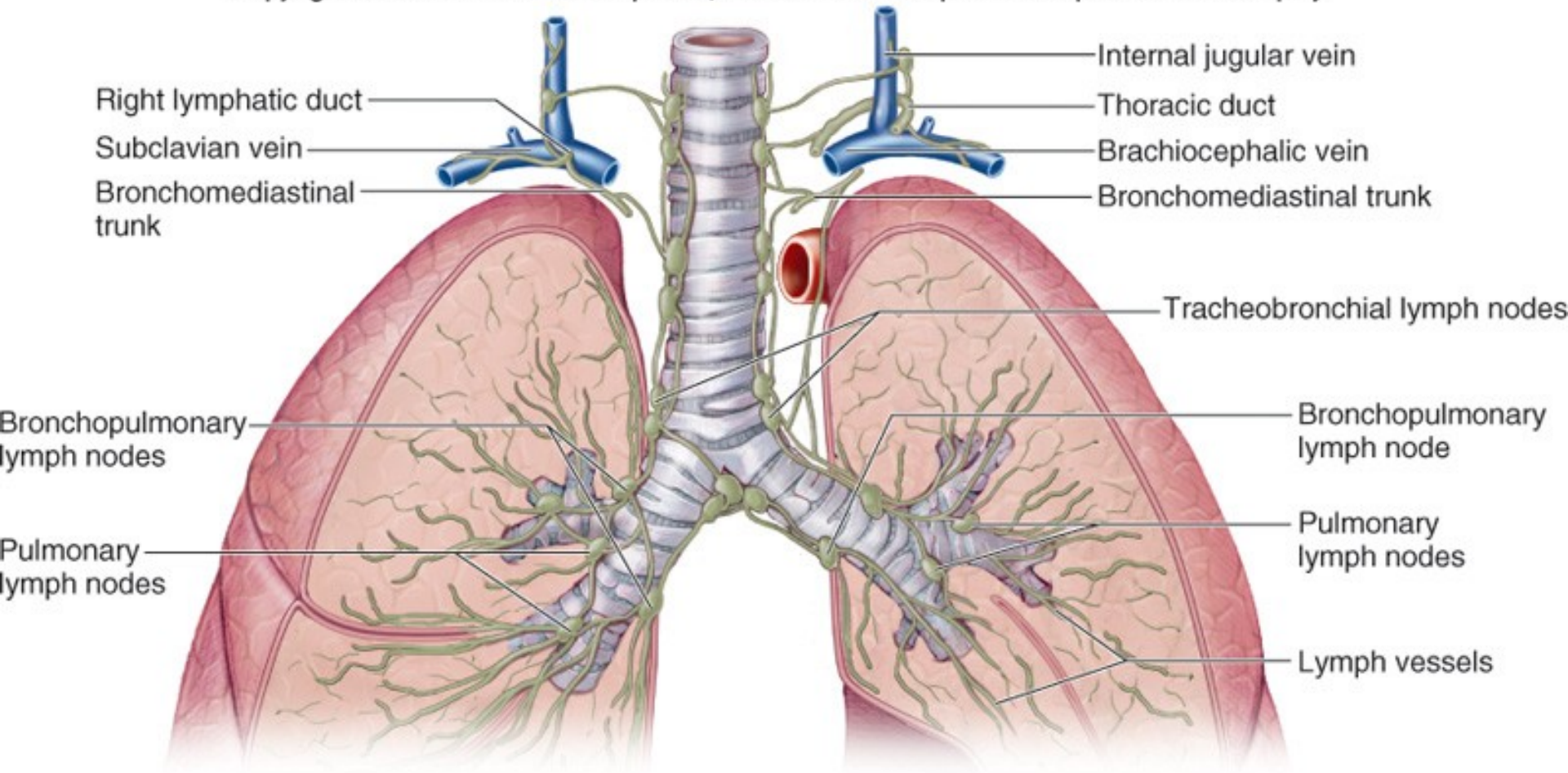
Blood Supply To and From the Lungs

- **Pulmonary trunk** and its branches carry deoxygenated blood
- **Bronchial arteries** bring oxygenated blood from the systemic circuit
- Two blood supplies **communicate** and most of the blood returns to the heart through the pulmonary veins

Lymphatic Drainage

- **Lymph nodes** and vessels are located within the connective tissue of the lung as well as around the bronchi and pleura.
- The lymph nodes collect carbon, dust particles, and pollutants that were not filtered out by the pseudostratified ciliated columnar epithelium.

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.



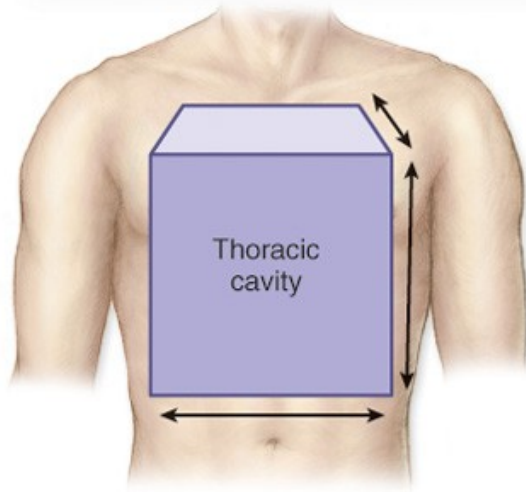
Thoracic Wall Dimensional Changes During Respiration

- Lateral dimensional changes occur with rib movements.
- **Elevation** of the ribs increases the lateral dimensions of the thoracic cavity, while **depression** of the ribs decreases the lateral dimensions of the thoracic cavity.

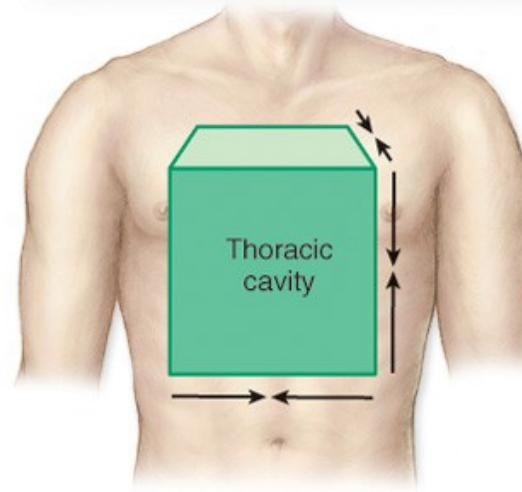
Muscles that Move the Ribs

- The ribs **elevate** upon contraction of the external intercostals, thereby increasing the transverse dimensions of the thoracic cavity during inhalation.
- Contraction of the internal intercostals **depresses** the ribs, but this only occurs during **forced** exhalation.
- **Normal** exhalation requires no active muscular effort.
- The **serratus posterior superior** elevates ribs 2–5 during inhalation, and the **serratus posterior inferior** depresses ribs 8–12 during exhalation.
- The **pectoralis minor**, **serratus anterior**, and **sternocleidomastoid** help with forced inhalation, while the **abdominal muscles** (external and internal obliques, transversus abdominis, and rectus abdominis) assist in active exhalation.

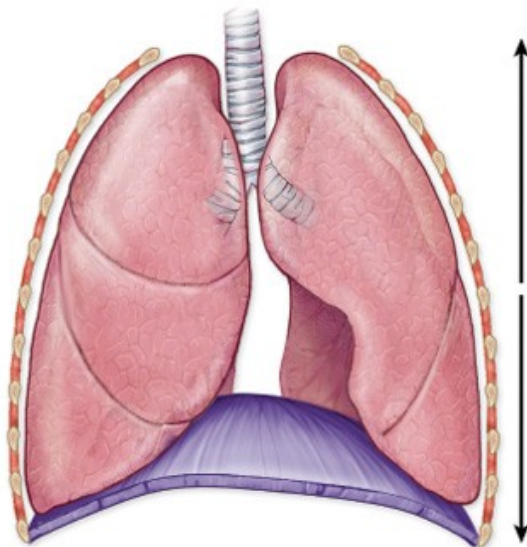
Inhalation



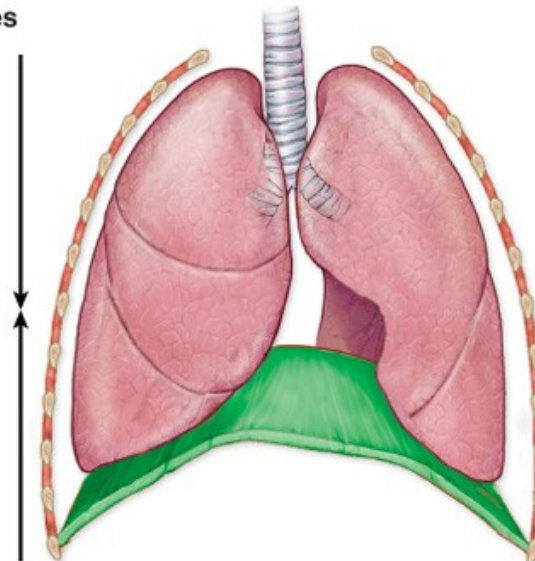
Exhalation



Vertical changes



Diaphragm contracts; vertical dimensions of thoracic cavity increase.



Diaphragm relaxes; vertical dimensions of thoracic cavity narrow.

Asthma

- **A chronic condition characterized by episodes of bronchoconstriction and wheezing, coughing, shortness of breath and excess pulmonary mucus.**
- Usually a sensitivity to an airborne agent. Upon reexposure, a localized immune reaction occurs in the bronchi and bronchioles.
- Usually lasts an hour or two.
- Eventually, the walls may become permanently thickened. Severe attacks may be fatal.
- Treatment: inhaled steroids and bronchodilators. Important to avoid triggering agent. Allergy shots may help some.